

Contact Transition Stability in the Impedance Control

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ABSTRACT

Contact stability has been considered based on simplified linearized models of robot and environment. It is shown that the most crucial parameter for contact stability is the target damping ratio. When a high compliance is required, the stability boundary conditions require a significantly overdamped impedance behavior to ensure a stable contact with a stiff environment. These conditions for a SISO system are defined in the analytic form. Obtained minimal damping ratio values are significantly lower than those previously proposed in the literature.

1 INTRODUCTION

The type of contact tasks may vary substantially depending on specific requirements, but the robot always has to perform three kinds of motion when realizing a contact task: so called gross motion, related to robot movement in free-space (free-motion mode), compliant (or fine, i.e. constrained) motion, related to robot movement constrained by an environment, and an interface transition motion, representing all passing-phases between free and compliant motion.

Many experimental investigations have demonstrated that the transition problem is very critical for most of the proposed compliant motion control algorithms, especially when the contact between end-effector and environment is stiff. However, the investigations were primarily concerned with the question of coupled stability (i.e. will the robot remain stable when it is interconnected with the environment ?) under various control algorithms, while assuming the manipulator initially is and always remains in contact with the environment. Surprisingly, relatively little research has addressed the problem of contact transition stability (i.e. will the robot, during transition from free- to contact-motion, establish a continuous contact with the environment without multiple impacts?) which is most fundamental for performing contact tasks in the presence of both unilateral (i.e. force) and bilateral (geometric) constraints.

Contact stability issues have been focused on in several studies, but relatively few have specifically addressed the transition stability in impedance control. The stability conditions have been investigated by Kazerooni *et al.* [1]. Using unstructured and structured models of the robotic manipulator and environment, the authors defined a general stability criterion in terms of operator norms. Colgate and Hogan [2] have demonstrated that the robot control system must present a passive impedance in order to remain stable in contact with an arbitrary passive environment. In [3], this result is extended to "nearly passive" systems, which are common in robotic practice. Many theoretical and experimental investigations have confirmed that high target damping and inertia are necessary to hold an impedance controlled robot against a surface [4]. Based on the equivalency between the force and impedance control, Volpe and Khosla [5] have recently established the transition stability condition for the impedance control imposing mass ratio (robot inertia through target inertia) to be less than one. But similar to [1], these results obtained on direct drive robots are not applicable to industrial

robots having very high equivalent Cartesian inertia (typically well over 100 kg) and the Cartesian position control stiffness of the order 10^6 N/m.

The equivalence between impedance and force control offers the possibility to apply the results on transition stability in force control [6] to the impedance control. However, in contrast to the impedance control algorithms, which provide the same control structure for the three motion phases, in force control schemes the transition to and from contact motion is usually based on discontinuous control (control switching) which is complex to implement in industrial robotic systems.

The control synthesis in the majority of the approaches is based on the increase of damping (a dominant real pole). However, the high values of target mass and damping needed to maintain a stable contact, give the robot a sluggish behavior. It is not desirable for handling both unexpected impact or loss of contact, resulting in high forces or long transition time, respectively. The established criteria are either necessary or sufficient, and usually quite conservative.

The purpose of this paper is to derive more reliable transition stability criteria pertinent for industrial robots and real contact tasks. Therefore, the consideration in this paper is mainly concerned with position-based impedance control, but under some circumstances the results can be extended to any controller realizing the target impedance. From this view, this work represents a continuation of previous investigations [7, 8]. Our aim is to examine the effects of impedance control parameters on the transition stability, rather than to develop complex transition control algorithms. The validity of derived contact stability conditions and control design is demonstrated in experiments with an industrial robot (Manutec-r3).

2 MATHEMATICAL MODEL

Here we should briefly consider the models of the robot constrained motion to be used for the design of the impedance control and consideration of the contact stability problem. Neglecting non-linear Coriolis and centrifugal effects due to relatively low operating velocities during contact and assuming the gravitational effect to be ideally compensated for, we obtain the linearized model around a nominal trajectory in Cartesian space $\mathbf{x}_0$ in the form

$$\mathbf{A}(\mathbf{x}_0)\ddot{\mathbf{x}} + \mathbf{B}(\mathbf{x}_0)\dot{\mathbf{x}} = \boldsymbol{\tau}(\mathbf{x}_0) + \mathbf{F} \quad (1)$$

$$\mathbf{A}(\mathbf{x}) = \mathbf{J}(\mathbf{q})^{-T} \bar{\mathbf{H}}(\mathbf{q}) \mathbf{J}(\mathbf{q})^{-1}; \mathbf{B}(\mathbf{x}) = \mathbf{J}(\mathbf{q})^{-T} \mathbf{b} \mathbf{J}(\mathbf{q})^{-1}$$

$$\boldsymbol{\tau} = \mathbf{J}(\mathbf{q})^{-T} \boldsymbol{\tau}_q; \bar{\mathbf{H}}(\mathbf{q}) = \mathbf{H}(\mathbf{q}) + \mathbf{H}_m$$

$$\mathbf{H}_m = \text{diag}(n_i^2 J_{mi}); \mathbf{b} = \text{diag}(n_i^2 b_{mi})$$

where: $\mathbf{q}$ -6x1 vector of the robot joint angles, $\mathbf{H}(\mathbf{q})$ -6x6 inertia matrix, n_i gear ratio, J_{mi} inertia of the motor actuator, b_{mi} viscous friction coefficient, i - denotes the local subsystems, $\boldsymbol{\tau}_q$ is a 6x1 vector of the control output torque reduced at the manipulator joints, $\mathbf{F}$ -6x1 vector of the Cartesian reaction contact (process) forces acting upon the end-effector, $\mathbf{J}(\mathbf{q})$ -6x6 Jacobian matrix which relates the end-effector velocities and

joint velocities (i.e. Cartesian forces and joint torques). In industrial robots, the Coulomb friction/stiction also plays an important role at low velocities and will be considered later as a disturbance to the linearized model (1).

In passive environments, it is convenient to adopt the relationship between forces and motion around the contact point in a linear form

$$-F = M_e \ddot{p} + B_e \dot{p} + K_e p \quad (2)$$

where $p = x - x_e$ is the robot *end-point penetration* into the environment, x_e is contact point locations, M_e , B_e , and K_e are respectively inertial, damping and stiffness matrices. Analogously the position difference $p_0 = x_0 - x_e$, which would be taken to mean a position planning failure due to tolerances or a desired penetration in the environment, will be referred to as *nominal penetration*. In the following we will consider contact with a surface and for the sake of simplicity, only the dominating stiffness effect will be taken into account.

3 POSITION BASED IMPEDANCE CONTROL

In commercially applied robotic systems it is most promising to implement position based compliance control schemes by closing force sensing-loops around the stable and robust position controller [7].

The position control system is based on the standardized decentralized position servo regulators. The position control law can be expressed in Cartesian space in the form

$$\tau = K_p(x_0 - x) - K_v \dot{x} + K_I \int (x_0 - x) dt \quad (3)$$

where the Cartesian and joint gains are related to each other as $K_p = J^{-1} k_p J^{-1}$ and analogous velocity and integral gains K_v and K_I , $k_p = \text{diag}(k_{p_i})$ represents the diagonal matrix of the local equivalent position feedback gains including scaling, transmission ratios, etc. Substituting (3) in (1) one obtains the closed loop transfer function of the position control system in the free-space

$$x = G_p(s)x_0; G_p(s) = [G_s(s)^{-1} + G_r(s)]^{-1} G_r(s) \quad (4)$$

where $G_r(s)$ and $G_s(s)$ are transfer functions of the regulator and robotic plant (i.e. robotic mechanical part) respectively

$$G_r(s) = Z_r = K_p + \frac{K_I}{s}; G_s(s)^{-1} = Z_s = \Lambda s^2 + (K_v + B)s \quad (5)$$

where the symbols Z_r and Z_s are introduced according to the means of these functions which represent impedances. Based on the above relations the closed-loop behavior of the robot arm in contact with a working environment (Fig. 1) can be expressed as

$$\begin{aligned} x &= G_{pE}(s)x_0 + G_{xE}(s)x_E \\ G_{pE}(s) &= [G_s(s)^{-1} + G_r(s) + G_E(s)]^{-1} G_r(s); \\ G_{xE}(s) &= [G_s(s)^{-1} + G_r(s) + G_E(s)]^{-1} G_E(s) \end{aligned} \quad (6)$$

The impedance control provides a general approach to the compliant motion control. The key issue in this approach is to control the relationship between the force acting on the end-effector and the error between nominal and actual positions of end-effector according to the law

$$-F = M_t \ddot{e} + B_t \dot{e} + K_t e \quad (7)$$

where $e = x_0 - x$ is position error (i.e. deviation between nominal and actual position), M_t and B_t and K_t are positive definite diagonal inertia, damping and stiffness matrices, respectively, which define the target impedance model.

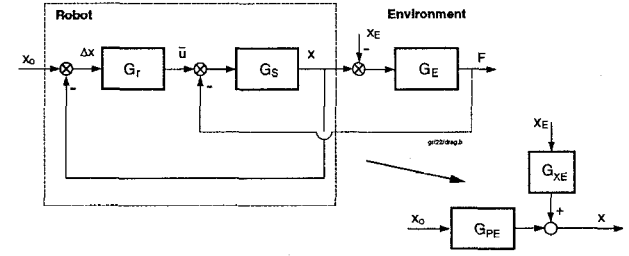


Figure 1 Position control system in contact with environment

The basic impedance control scheme corresponding to the considering linearized robot position control is depicted in (Fig. 2). This scheme consists of two control loops: an inner position servo and an outer loop generating position command modification Δx_F based on end effector force/torque measurements. The outer loop is naturally closed when the end-effector encounters the environment.

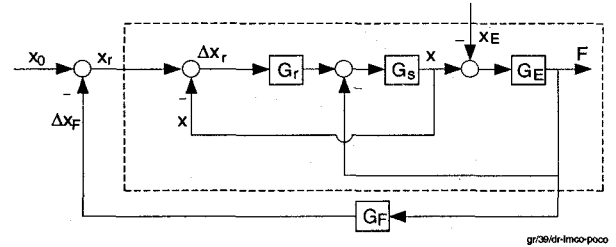


Figure 2 Basic impedance control scheme

The entire impedance control design problem is split into two subproblems: realization of the target impedance model; and design of the target impedance parameters to meet a stable interaction with environment and required performance.

As demonstrated in [8], the performance of commercial industrial robots systems allow the significant simplification of the impedance control design and implementation. The robustness of internal position control allow the disturbances due to interaction force and joint friction effects to be neglected. Furthermore, due to high gear ratios and accurate design of joint position controller, the closed loop position control transfer matrix G_p is *diagonally dominant* and *spatially rounded* with good approximation, i.e. it exhibits similar performances independent of Cartesian directions or compliance frame selection, and, far from singularities, has similar performance in a large workspace area. These characteristics, highly desired in position control to guarantee good and uniform trajectory tracking independent of motion direction nor robot position, allow the impedance control to be implemented in the same spirit as the position controller. Thus, we obtain a simple and reliable impedance control compensator in the frequency domain

$$G_F(s) = \hat{G}_p(s)^{-1} G_t(s)^{-1} \quad (8)$$

where G_t is the diagonal target end-effector impedance matrix specifying the target behavior in each compliance frame directions corresponding to (6), G_p is the diagonal estimate of the closed loop position transfer matrix. Based on (4-8) the above control law provides the following nominal closed loop

contact behavior

$$\mathbf{x} = \mathbf{G}_p(s)\mathbf{x}_0 + \mathbf{G}_t(s)^{-1}\mathbf{F} \quad (9)$$

Substituting (7) in (9) yields

$$\mathbf{G}_t(s)\mathbf{e}(s) = \mathbf{F}(s) + \mathbf{G}_t(s)(\mathbf{I} - \mathbf{G}_p(s))\mathbf{x}_0(s) \approx \mathbf{F}(s) = \mathbf{G}_e(s)\mathbf{p}(s) \quad (10)$$

corresponding to the target model. Stiff position controller causes the nominal motion perturbation term in (10) to vanish in a relatively short period. For the analysis of the force history during contact, it is very useful to express the contact force in terms of nominal penetration

$$\mathbf{F}(s) = [\mathbf{G}_t(s)^{-1} + \mathbf{G}_e(s)^{-1}]^{-1} \mathbf{G}_p(s)\mathbf{p}_0(s) = \mathbf{Z}_C(s)\mathbf{p}_0(s) \quad (11)$$

where the dimension and physical meaning of the transfer function corresponds to an impedance referred to as *coupled impedance* in [8]. The realization of the target model by the proposed control law has been demonstrated in [7], where experiment and simulation within system bandwidth is relatively well-matched

4 CONTACT STABILITY CRITERIA

The objective is here to define necessary and sufficient conditions for the stable transition in the impedance control. From a pragmatic viewpoint, the contact transition can be considered as stable, if the contact is not lost after the manipulator hits the environment. A stable contact transition can be characterized by non-zero force (after contact is detected), positive penetration of manipulator end point into environment, nonappearance of bouncing, etc. The most critical issue in transition control is the initial impact against a stiff environment. Obviously a stable controller should ensure the passage through the transition phase with maintaining contact until all impact energy has been absorbed.

The contact stability condition imposes that the actual penetration $\mathbf{p}$ is positive during contact, i.e.

$$|\mathbf{x}(t; t_0, \mathbf{x}_{t_0})| \geq |\mathbf{x}_e| \quad (12)$$

in each contact direction for $t \geq t_0$, where t_0 denotes the time instance when the first contact $\mathbf{x}_{t_0} = \mathbf{x}_e$ is established. An equivalent criterion by a suitable orientation of the compliance frame, relative to which the target model and contact force are defined, can be expressed by $\mathbf{F}(t) \geq 0$.

Another contact stability condition is based on the simple geometric consideration (therefore we refer it to as *geometric criterion*) and is expressed by

$$\frac{|\mathbf{e}(t)|}{|\mathbf{p}_0(t)|} = \frac{|\mathbf{p}_0(t) - \mathbf{p}(t)|}{|\mathbf{p}_0(t)|} = \frac{|\mathbf{x}_0(t) - \mathbf{x}(t)|}{|\mathbf{x}_0(t) - \mathbf{x}_e|} \leq 1 \quad (13)$$

This criterion presumes the position deviation to be smaller or equal to the nominal penetration for $t \geq t_0$. In other words, the above relation implies the actual end-effector position during a stable contact to always be located between the position of environment and the nominal position. Indeed, this simple geometrical criterion cannot be considered as a global contact stability condition. However, in a contact between industrial robotic systems with very stiff position servos and a relatively rigid environment, where impedances are significantly higher than the target one, it is, as will be shown, quite correct and convenient to apply the assumption (13). Moreover, the relative criterion is well suited to specify and analyze the transition problem in terms of input-output stability. The

stability problem at the contact between a stiff robot and an elastic environment, where in general (13) is not valid, is not critical.

Here we will examine the contact transition problem during an elemental positioning task in which a nominal position behind a hindrance is commanded and a stable transition and equilibrium should be realized (Fig. 3). Dependent on the way the bounds of variables and parameters characterizing the contact transition (actual and nominal penetration, and target impedance parameters) are estimated, the following stability analysis methods are established.

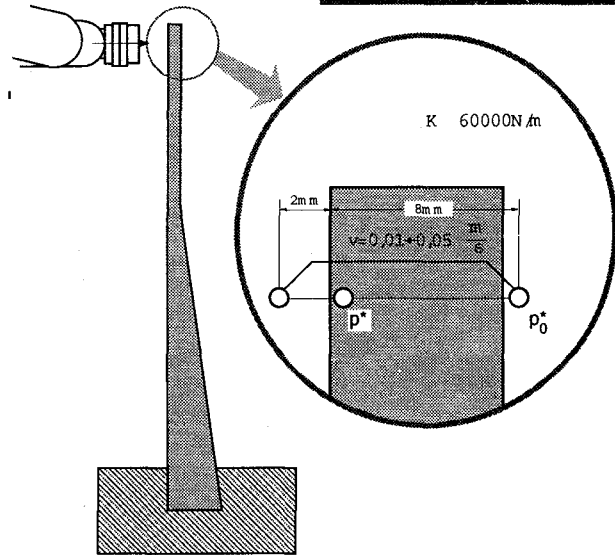
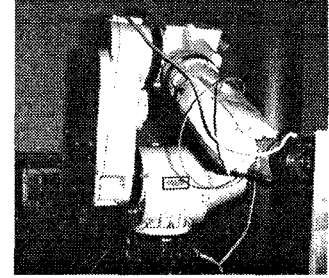


Figure 3 Basic contact task

4.1 Time domain criteria for stability

A good place to start the analysis is the ideal target impedance model (10) realized by the control law (8) in each constrained direction. Using (2), the target impedance model can be rewritten as (for the sake of simplicity the environment is presented as a stiffness)

$$\ddot{\mathbf{e}} + 2\xi_t\omega_t\dot{\mathbf{e}} + (\omega_t^2 + \omega_e^2)\mathbf{e} = \omega_e^2\mathbf{p}_0 \quad (14)$$

where the nominal trajectory $\mathbf{p}_0(t)$ is a continuous and bounded function, with initial condition $\mathbf{p}_0(t_0) = 0$, and $\mathbf{p}_0 \rightarrow \mathbf{p}_0^*$ as t increases, where $\mathbf{p}_0^*$ is the maximum penetration corresponding to the nominal goal position (Fig. 3); $\xi_t = B_t/2\sqrt{K_t M_t}$ and $\omega_t^2 = K_t/M_t$, $\omega_e^2 = K_e/M_t$. Under these circumstances, the position deviation $\mathbf{e}(t)$ and actual penetration $\mathbf{p}(t)$ reach the equilibriums when $t \rightarrow \infty$

$$\mathbf{p}^* = \frac{1}{1+\kappa}\mathbf{p}_0^*; \mathbf{e}^* = \frac{\kappa}{1+\kappa}\mathbf{p}_0^* \quad (15)$$

where κ denotes stiffness ratio $\kappa = K_e/K_t \gg 1$. As well

known, in the second order perturbed system (14) these equilibriums are for $\xi_t > 0$ (ensuring negative real part of characteristic equation eigenvalues) asymptotically stable. This result can be established by considering the equivalent system obtained by shifting the origins $\bar{e} = e - e^*$, i.e. $\bar{p} = p - p^*$,

$$\ddot{\bar{e}} + 2\xi_t \omega_t \dot{\bar{e}} + (\omega_t^2 + \omega_e^2) \bar{e} = \omega_e^2 (p_0 - p_0^*) \quad (16)$$

and corresponding Liapunov function

$$V = \frac{1}{2} [\dot{\bar{e}}^2 + (\omega_t^2 + \omega_e^2) \bar{e}^2]; \dot{V} = -2\xi_t \omega_t \dot{\bar{e}}^2 + \dot{\bar{e}} \omega_e^2 (p_0 - p_0^*) \leq 0 \quad (17)$$

A typical transition behavior in the penetration phase plane is sketched in (Fig. 4). For the positive nominal velocity during nominal motion phase $p_0 > 0$, the deviation velocity is usually also positive, thus the system is globally asymptotically stable. However, the estimates of contact transition measures (12, 13) based on the Liapunov functions (17) might be very conservative and useless for the transition stability analysis. Obviously, for the considered perturbed linear system more accurate bounds can be obtained by solving the equation (14). Assuming the initial condition at the contact $e(t_0) = \dot{e}(t_0) = 0$, which corresponds to an ideal position servo, the solution of differential equation (14) for the under critical damping case $\xi_t < \sqrt{1+\kappa}$ (we are looking for the minimum damping ensuring a stable contact) has the form

$$e = e_p = \frac{\omega_e^2}{\omega_d} \int_{t_0}^t p_0(\zeta) e^{-\xi_t \omega_t (t-\zeta)} \sin \omega_d (t-\zeta) d\zeta \quad (18)$$

where $\omega_d = \omega_t \sqrt{1+\kappa - \xi_t^2}$. Since the nominal penetration function is characterized by standard velocity and acceleration profiles, it is interesting to analyze their influence on the contact stability. Assuming $p_0(t)$ and its derivative to be a continuous function, we can express the position deviation and penetration in terms of nominal motion functions by means of partial integration

$$\begin{aligned} e(t) = & \frac{\kappa}{1+\kappa} \left(p_0(t) - \frac{2\kappa}{\omega_t(1+\kappa)} \dot{p}_0(t) + \frac{\hat{v}_0}{\omega_d} e^{-\xi_t \omega_t (t-t_0)} \sin[\omega_d(t-t_0) - 2\theta] \right) \\ & - \frac{1+\kappa - 4\xi_t^2}{\omega_t^2(1+\kappa)^2} \ddot{p}_0(t) + \frac{\hat{a}_0}{\omega_t \omega_d \sqrt{1+\kappa}} e^{-\xi_t \omega_t (t-t_0)} \sin[\omega_d(t-t_0) - 3\theta] \\ & + \frac{1}{\omega_t^2 \sqrt{1+\kappa - \xi_t^2}} \int_{t_0}^t \ddot{p}_0(\zeta) e^{-\xi_t \omega_t (t-\zeta)} \sin[\omega_d(t-\zeta) - 3\theta] d\zeta \end{aligned} \quad (19)$$

where

$$\pi/2 \leq \theta \leq \pi := \sin \theta = \frac{\sqrt{1+\kappa - \xi_t^2}}{\sqrt{1+\kappa}}; \cos \theta = \frac{-\xi_t}{\sqrt{1+\kappa}} \quad (20)$$

and $\hat{v}_0 = \dot{p}_0(t_0)$, $\hat{a}_0 = \ddot{p}_0(t_0)$. If the higher nominal motion derivatives are discontinuous, the transition can be analyzed within motion segments $p_0(t) \ t \in [t_1, t_2]$, based on (19), whereby well known homogenous solution should be added to the particular one (18), dependent on initial conditions at each segment.

The above equations provide a useful base to analyze the transition behavior during the entire motion or particular nominal path segments. Thereby, the following might be observed: a) in the convenient target impedance model, the

behavior of penetration, i.e. motion deviation and force, is significantly shaped by the nominal motion; b) a positive nominal acceleration at contact (i.e. increasing velocity), as well as increasing damping ratios and target frequency, support a stable transition.

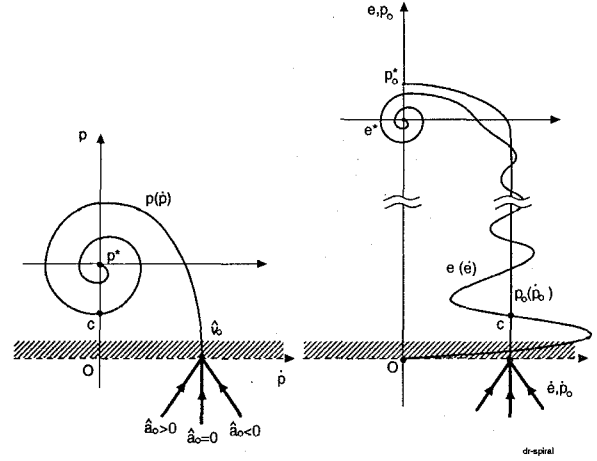


Figure 4 Typical contact transition behavior

The critical point C for the loss of contact in the $(\dot{p}, p)$ phase plane (Fig. 4) can be determined by differentiating (19) and setting $\dot{e} = \dot{p}_0$, which corresponds to $\dot{p} = 0$. The obtained equation, together with (12, 13) and (19), defines necessary and sufficient conditions ensuring a stable contact transition. However, using these equations for control synthesis is quite complex (expressed by systems of transcendental algebraic equations) even in the considered elemental case. Usually some of the target impedance parameters are prescribed by the control task, such as target stiffness determining the maximum steady-state force for an estimated tolerance (maximum nominal penetration) based on (11). The maximum target frequency is limited by the position control bandwidth ω_p (usually $\omega_t \leq 0.5\omega_p$). Thus the above defined stability conditions allow us to determine the target damping ensuring a stable contact transition. For example, assuming the trapezoidal nominal velocity profile, and the transition during the constant velocity phase (Fig. 4), the condition $\dot{p} = 0$ yields

$$e^{-\xi_t \omega_t t} \sin(\omega_d t - \theta) = \frac{\sin \theta}{\kappa} \quad (21)$$

Instead of solving the system of transcendental equations (21), (19) and (13), we can from the above equation estimate the time period needed to reach the critical point C (Fig. 4) $(\pi + \theta)/\omega_d < t_C < (3\pi/2 + \theta)/\omega_d$. Substituting the lower value in (19) and (13), that provides more safety result, yields

$$e^{\frac{\pi + \theta}{\omega_d}} \leq \frac{\pi + \theta}{\kappa \sin \theta} - 2 \cos \theta \quad (22)$$

The solution of the above transcendental equation within the interval (20), determines the minimum value of target damping ratio ξ_t ensuring the stable contact transition. The interesting result is that in the considered ideal case, the bound of contact stability is influenced by target damping ξ_t and stiffness ratios κ , but it is independent of the frequency ω_t .

5.2 Influence of Coulomb friction and force time lags

The influence of the Coulomb friction on the contact transient process can be analyzed by introducing the equivalent

Cartesian friction force in the considered impedance control model

$$\mathbf{G}_t(s)\mathbf{e}(s) = \mathbf{F}(s) + \mathbf{G}_t(s)(\mathbf{I} - \mathbf{G}_p(s))\mathbf{x}_0(s) + \mathbf{S}_p(s)\mathbf{G}_s(s)\mathbf{F}_c(s) \quad (23)$$

where $\mathbf{F}_c$ is Coulomb friction force, while $\mathbf{S}_p(s)$ denotes the position control sensitivity transfer function. Since in industrial robotic systems with very stiff position controllers, the sensitivity function is designed to be very small within the position control (as well as position based impedance control) bandwidth, it might be concluded that the influence of $\mathbf{F}_c$ on the transition process is negligible. Moreover, considering the motion resistant effect of the Coulomb friction, $\mathbf{F}_c$ supports the end-effector to remain in the contact. Thus, the stability results obtained with the ideal model are on the safe side.

Quite opposite is the effect of control time lags. The difference-differential equation describing the closed loop behavior of the proposed control with retarded force information can be presented in the form

$$\ddot{\mathbf{e}}(t) + 2\xi_t\omega_t\dot{\mathbf{e}}(t) + \omega_t^2\mathbf{e}(t) + \omega_e^2\mathbf{e}(t-\tau) = \omega_e^2\mathbf{p}_0(t) \quad (24)$$

where τ is time delay. A necessary and sufficient condition for the asymptotic stability of the solution of the stationary equation is negativity of the real parts of all the roots of the corresponding characteristic quasipolynomial

$$z^2 + 2\xi_t\omega_t z + \omega_t^2 z + \omega_e^2 e^{-\tau z} = 0 \quad (25)$$

In order to analyze the contact stability of the retarded system based on previous results, it is suitable to utilize the equivalency between the retarded system (24) and the ideal one (14) (see [8] for more details). For the retarded second order system (24) it is possible to find an equivalent system without retardation having the same fundamental solutions. A numerical procedure designed to compute the contact stability bounds of the retarded system based on the second order system (14) is presented in [8]. In general, the retarded system requires a significantly higher amount of damping in order to stabilize the transition process with delayed force signals. A further point of interest is that the target frequencies of the retarded and equivalent systems (14) are different, and thus the frequency becomes very important for the transition stability in the real delayed systems. In principle, higher target frequency (lower mass) is more suitable to cope with the delay problem during transition.

5.3 Input/output stability criteria

The contact stability condition imposes that the actual penetration $\mathbf{p}$ is positive during contact. Assuming that the nominal penetration $\mathbf{p}_0(t)$ is always positive, the coupled impedance transfer function (11) has to be a positive (i.e. strictly positive) real function (SPR). SPR functions describe behavior of passive mechanical systems or networks. System passivity means that the work is always done by the external force, without loss of the contact. The simplified coupled impedance transfer function obtained from (11) for an ideal position servo $\mathbf{G}_p = 1$, describes a passive system and can be applied to contact stability analysis based on the positivity criterion. Since both target and environment transfer functions are assumed to be Hurwitz polynomials, for contact stability it is necessary that in each SISO system along constrained frame directions

$$\operatorname{Re}\left\{\mathbf{G}_t(s)/[\mathbf{G}_t(s) + \mathbf{G}_e(s)]\right\} \geq 0 \quad s = j\omega; \quad \omega \in [0, \infty] \quad (26)$$

is fulfilled.

This criterion yields in a pure stiffness environment

$$\xi_t \geq 0.5(\sqrt{1+\kappa} - 1) \quad (27)$$

Another possibility to define stable contact condition provides the infinity norm of the relationship (13), which based on (11) yields

$$\sup \frac{\|\mathbf{e}(t)\|_2}{\|\mathbf{p}_0(t)\|_2} < \|\mathbf{G}\|_\infty = \sup_\omega \bar{\sigma}\left(\mathbf{G}_p(j\omega) \frac{\mathbf{G}_e(j\omega)}{\mathbf{G}_e(j\omega) + \mathbf{G}_t(j\omega)}\right) \leq 1 \quad (28)$$

where $\bar{\sigma}$ denotes the largest singular value. Since in the convenient position control systems position control norm is less than one (small gain theorem), it follows

$$\|\mathbf{G}_e(j\omega)/(\mathbf{G}_e(j\omega) + \mathbf{G}_t(j\omega))\|_\infty \leq 1 \quad (29)$$

Along each compliance frame direction this implies for the considered target impedance model

$$\xi_t \geq 0.5(\sqrt{1+2\kappa} - 1) \quad (30)$$

The physical meaning of the criterion (28) is that the energy of the target model is less than the input nominal motion energy (the difference corresponds to the absorbed energy during transition). The last condition requires somewhat higher damping coefficients to ensure contact stability in comparison with (27). In spite of an effective and simple formulation, this criterion defines only a sufficient condition. However, due to the conservative infinity norm the obtained results are on the safe side. It should be mentioned, that the damping ratio bound (28) is still less than the critical damping value proposed by Benady et al. [4] imposing $\xi_t \geq \sqrt{1+\kappa}$.

A very important advantage of the input/output criterion (28) is that it can be applied for the continuous as well as for the discrete systems including the time lags (25). The weighting functions representing the model uncertainties can also be included. Figure 5 illustrates the derived contact stability limits for the ideal target impedance model without retardation.

5 EXPERIMENTAL TESTS

In order to test obtained contact stability conditions, the experiments were performed on the Manutec-r3 robot using the IPK's experimental control system. The proposed control scheme is implemented with a computational cycle of 8 ms with a maximum force sensing and computation time lag of 8 ms. In the experiments a nominal goal position 8 mm beyond the contacted surface is commanded (Fig. 3). The target stiffness of $\mathbf{K}_t$ is varied in the interval [250, 500, 1000, 2000, 4000 8000] N/m, while the target frequencies are settled to [0.5, 1, 3, 6, 8] Hz respectively, by setting the target mass for each above target stiffness in the interval [0.1, ..., 200] kg. Damping ratios bounds are computed using the following criteria a) time continuous system with retardation (24) b) infinity norm criterion (28) including the delay 8ms in the denominator (25) and c) the middle value between these two bounds. The stiffness of the environment was $\mathbf{K}_e = 60000$ N/m in the experiments (Fig 3).

For the each set of parameters a contact transition experiment has been performed. A typical result is shown in (Fig. 6) presenting the contact force trajectory in experiments with variable damping ratios. The estimated real contact bounds obtained in experiments are illustrated in (Fig. 7). From the experiments it can be observed that the contact stability limits obtained for the simple second order system with delay

provides quite accurate values for the high target frequencies (6-8 Hz) and relatively low stiffness ratios. The target frequency 8 Hz is near to the position control bandwidth limits, and should be avoided in real applications. At lower ω_t (0.5-1 Hz), more accurate transition is achieved at higher damping values. The infinity norm criterion (28) provides in all cases higher damping values than other methods, but due to simple computations it is quite practicable, especially to cover model uncertainties and disturbances. Finally, the experiments have proven the contact stability analysis obtained on the simple linearized target impedance model.

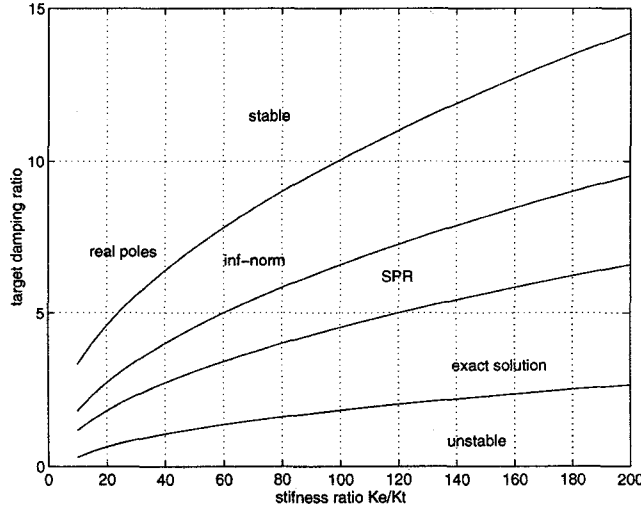


Figure 5 Contact stability bounds

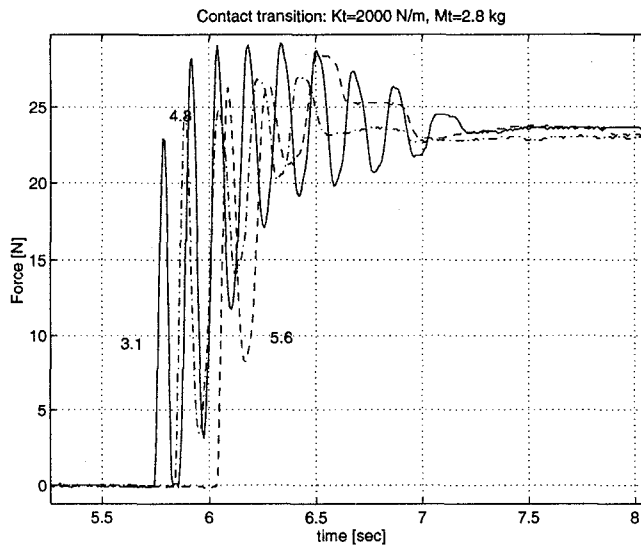


Figure 6 Contact force transitions for various ξ_t ($\omega_t = 6\text{Hz}$)

6 CONCLUSION

This paper considers the contact stability problem which was recently found to be fundamental for performing almost all contact tasks in advanced robot applications. It is shown that the most crucial parameter for contact stability is the target damping ratio. When a high compliance is required, the stability boundary conditions require a significantly overdamped impedance behavior to ensure stable contact with a stiff environment. These conditions for a SISO system are

defined in the analytic form. Obtained minimal damping ratio values are significantly lower than those previously proposed in the literature. They result in a faster system response and lower force overshoots during transitions. The time delay in the real control system is the crucial influence on the lost of contact. The experiments have proven the obtained conditions which guarantee a stable contact transition in real robotic systems.

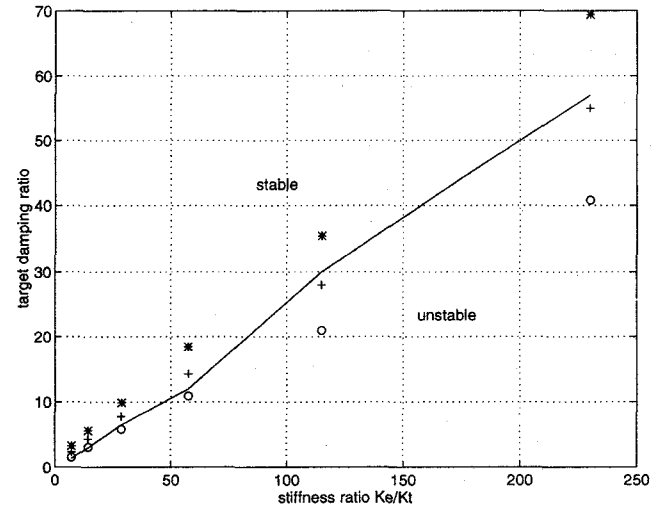


Figure 7 Experimental stability bounds ($\omega_t = 6\text{Hz}$, $t=8\text{ ms}$) retarded second order system 'o', infinity-norm '*', middle values '+', experiment 'solid'

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